

Solar Power System

Detailed Engineering Proposal & Quotation

SYSTEM SIZE

30 kWp

52 modules

STORAGE

40 kWh

BYD

BILL OFFSET

92%

Year 1

Prepared For

Client: Acme Industries Ltd
Site: Industrial Area, Nairobi
Phone: —

Company: —
Email: ops@acme.co.ke
Reference: ACM-2026-001

Document Control

Proposal #: SGP-2026-TEST
Valid until: 2026-05-21
Currency: KES

Issued: 2026-04-21
Prepared by: Eng. J. Doe
Standards: IEC 62548 · IEC 60364-7-712 · IEEE 1547

1. Executive Summary

Prepared for Acme Industries Ltd at Industrial Area, Nairobi, this proposal recommends a 30 kWp solar PV system with 40 kWh of lithium battery storage. Based on the site's measured solar resource of 5.6 kWh/m²/day (NASA POWER) and current monthly consumption of 4500 kWh, the system is engineered to offset approximately 92% of the load in Year 1 and pay for itself in 2.6 years.

Headline Numbers

System size:	30 kWp	Annual production:	48,600 kWh/yr
Specific yield:	1620 kWh/kWp/yr	Year-1 savings:	KES 1,180,000
25-yr lifetime savings:	KES 40,306,162	Capex:	KES 3,080,000
Simple payback:	2.6 yrs	IRR (20-yr):	31.4 %
NPV @ 10%:	KES 8,400,000	CO ₂ avoided (lifetime):	11720 mtCO ₂ e

2. Site Assessment

Latitude:	-1.286	Longitude:	36.817
Elevation:	1670 m	Climate zone:	Tropical highland
Solar resource:	5.6 kWh/m²/day	Monthly consumption:	4500 kWh/month
Peak demand:	28 kW	Current tariff:	KES 25.5/kWh
Tariff schedule:	CI1 (KPLC commercial)	Grid reliability:	~3 outages/month, 2-4h average

3. System Architecture

Topology:	Hybrid grid-tied + storage	DC capacity:	30 kWp
Module count:	52	Module type:	Jinko Tiger Neo 72HL4-BDV 580 W
Inverter:	Sungrow SG110CX 25 kW	Battery:	BYD Battery-Box Premium HVM 40 kWh
Tilt / Azimuth:	10° / 0°	String config:	2 strings x 13 modules
Specific yield:	1620 kWh/kWp/yr	System losses:	14%

3.1 Estimated Monthly Energy Yield (kWh)

Month	Production (kWh)	Cum. (kWh)
Jan	3,800	3,800
Feb	3,650	7,450
Mar	4,100	11,550
Apr	3,950	15,500
May	3,700	19,200
Jun	3,500	22,700
Jul	3,550	26,250
Aug	3,850	30,100
Sep	4,200	34,300
Oct	4,350	38,650
Nov	4,150	42,800
Dec	3,850	46,650

4. Bill of Materials & Pricing

#	Item	Qty	Unit (KES)	Total (KES)	Datasheet
1	Jinko Tiger Neo 580W	52	12,500	650,000	see appendix
2	Sungrow SG110CX 25kW	1	420,000	420,000	see appendix
3	BYD HVM 16.6kWh	2	580,000	1,160,000	see appendix
4	DC isolators + cabling	1	180,000	180,000	—
5	Mounting (galvanised steel)	1	250,000	250,000	—
6	Installation + commissioning	1	420,000	420,000	—
Subtotal (excl. VAT)				3,080,000	

5. Investment & Returns

Capital cost:	KES 3,080,000	Year-1 savings:	KES 1,180,000
Simple payback:	2.6 yrs	IRR (20-yr):	31.4%
NPV:	KES 8,400,000	Discount rate:	10%
Tariff escalation:	3% / yr	Panel degradation:	0.5% / yr

5.1 25-Year Projected Cash Flow

Year	Net cash (KES)	Cumulative (KES)
1	1,180,000	-1,900,000
2	1,209,500	-690,500
3	1,239,738	549,238
4	1,270,731	1,819,969
5	1,302,499	3,122,468
6	1,335,062	4,457,530
7	1,368,438	5,825,968
8	1,402,649	7,228,617
9	1,437,715	8,666,332
10	1,473,658	10,139,990
11	1,510,500	11,650,490
12	1,548,262	13,198,752
13	1,586,969	14,785,721
14	1,626,643	16,412,364
15	1,667,309	18,079,673
16	1,708,992	19,788,665
17	1,751,717	21,540,382
18	1,795,510	23,335,892
19	1,840,397	25,176,289
20	1,886,407	27,062,696
21	1,933,567	28,996,263
22	1,981,907	30,978,170
23	2,031,454	33,009,624
24	2,082,241	35,091,865
25	2,134,297	37,226,162

5.2 Financing Options

Option	Term (mo)	Rate (% APR)	Monthly (KES)	Total paid (KES)
Equity	0	0	0	3,080,000
Bank loan 5yr	60	14	72,000	4,320,000
Lease-to-own 7yr	84	16	68,000	5,712,000

6. Environmental Impact (lifetime)

Clean energy generated:	1180 MWh	C O , a v o i d e d :	7 2 0 t C O , e
Equivalent trees planted:	12000	Cars off the road (1 yr):	156

7. Warranty Matrix

Component	Product (yrs)	Performance (yrs)	Notes
Modules	15	30	Linear performance warranty
Inverter	10	-	+5yr extension available
Battery	10	-	60% retained capacity at 10yr / 6000 cycles
Workmanship	2	-	From commissioning

8. Scope of Work

Included

- Detailed design + permit pack
- Supply of all listed equipment
- Roof structural assessment
- DC/AC installation + commissioning
- 24-month warranty + remote monitoring

Excluded (client to provide)

- Civil works (slab/trenching) > KSh 50,000
- Utility interconnection fees
- Optional: SCADA / off-site monitoring portal

Deliverables

- Stamped single-line diagram
- As-built drawings
- O&M manual
- Test & commissioning certificate
- EPRA permit submission pack

8.1 Project Milestones

Stage	Days	Deliverable
Design freeze	7	Stamped drawings + BOM
Equipment delivery	35	Goods received note
Mechanical install	49	Roof + structure complete
Electrical + commissioning	56	Energised system
Handover	63	Acceptance certificate

9. Terms & Conditions

- 1. This quotation is valid until 2026-05-21. Pricing assumes current KES exchange rates and component availability.
- 2. Payment terms: 50% deposit on order, 40% on equipment delivery, 10% on commissioning. Bank details supplied on order acceptance.
- 3. Lead time: 4–6 weeks from cleared deposit, subject to import logistics.
- 4. Workmanship warranty: 24 months from commissioning. Equipment warranties as per the manufacturer matrix in §7.
- 5. Performance guarantee assumes the site conditions stated in §2 and no shading exceeding the modelled values.
- 6. All designs comply with IEC 62548 (PV array), IEC 60364-7-712 (PV systems), and IEEE 1547 (interconnection). Local AHJ permits are the contractor scope.
- 7. Disputes resolved by arbitration under the laws of the Republic of Kenya.

10. Client Acceptance

I, the undersigned, accept this proposal and authorise the works above.

Signature

Name: Acme Industries Ltd

Date

Title:

Appendix A · Data Sources

In line with our data-honesty policy, every figure in this proposal is traceable to an authoritative source. The table below lists every dataset, API or datasheet relied upon in this document.

Field / dataset	Source	URL / reference	Retrieved (UTC)
Solar irradiance	NASA POWER (LARC)	https://power.larc.nasa.gov	—
Sun position	Michalsky (1988) algorithm	—	—
POA irradiance	Hay & Davies (1980); Perez et al. (1990)	—	—
BOS losses	NREL SAM 12-component loss stack	https://sam.nrel.gov	—
Tariff schedule	EPRA Kenya retail tariff (current schedule)	https://www.epra.go.ke	—
Grid emission factor	IFI Dataset of Default Grid Factors v3.2 / IEA	https://www.ifi.org	—
Standards	IEC 62548, IEC 60364-7-712, IEEE 1547	—	—
Tariff KSh 25.5/kWh	EPRA CI1 Tariff Order Q1 2026	https://epra.go.ke	2026-04-21
Equipment pricing	EmersonEIMS supplier quotes	internal	2026-04-21

Estimates derived from the above sources are labelled "(estimate)" or "(regional est.)" wherever they appear above. No figure in this document has been fabricated or extrapolated without disclosure.